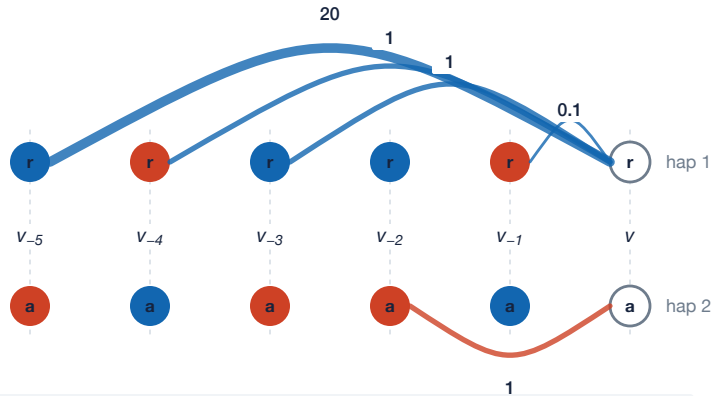


## a Weighted votes from up to k = 35 upstream variants



$$h_1(v) = 20 + 1 + 1 + 0.1 = 22.1$$

$$h_2(v) = 1$$

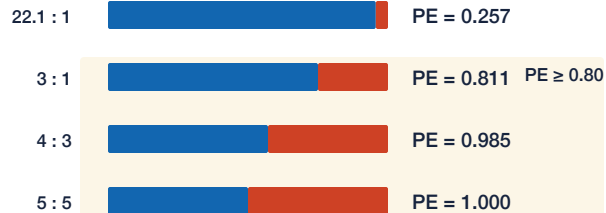
$v \rightarrow$  **haplotype 1** (a tie opens a new phase block)

## c Block formation



## b Phasing entropy

$$PE(v) = - \sum p_i \log_2 p_i, p_i = h_i / (h_1 + h_2)$$



$PE \geq 0.80 \rightarrow$  a GNN window is opened (Fig. 6)

written to VCF as INFO/PE, INFO/H1, INFO/H2

allele on haplotype 1

allele on haplotype 2